Vuppula Hemanth Reddy

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.70
Narayana Junior College,Adibatla	Class XII (Telangana State Board of Intermediate Education)	2023	98.10%
Narayana School,Karmanghat	Class X (Board of Secondary Education,Telangana state)	2021	10.00%

SKILLS

Languages: Python,C, C++,SQL

Libraries: NumPy, Pandas, Matplotlib, Scikit-learn,Pytorch

Tools: Git, LaTeX, Linux

Familar with: HTML, CSS, JavaScript,React,Node.js, Express.js,TensorFlow

RELEVANT COURSES

AI: Foundations of Machine Learning, Deep Learning, Online Learning, Computer Vision, Natural Language Processing

CS: Computer Architecture, Discrete Mathematics, DBMS, Data Structures and Applications, Algorithms, Introduction to Programming, Software Engineering

Mathematics: Probability and Random Variables, Matrix Theory, Calculus, Vector Calculus, Differential Equations, Applied Statistics

PROJECTS

Ventilation Failure Prediction in COPD ICU Patients *(Research Project / Dr.Rahuldeb Sarkar)*

- Developed XGBoost classification model to predict ventilation failure using clinical data from MIMIC-IV.
- Evaluated HybridGNet, TorchXRayVision PSPNet, CXAS, and IanPan for chest X-ray anatomical segmentation and radiological feature extraction.
- Extracted quantitative lung, cardiac, airway, and hilar features from CXR images, including lung area, radiodensity, asymmetry, cardiothoracic ratio, and boundary compactness.
- Integrated clinical and radiological features for ventilation-failure prediction, improving AUROC from 0.8482 to 0.9407 and AUPRC from 0.5177 to 0.7570.

Analytics BI *(Software Engineering / Group Project / Prof. Manish Singh)*

- Built a full-stack BI platform supporting data ingestion, visualization, and interactive dashboard management.
- Implemented CSV/XLS/XLSX ingestion pipelines with streaming parsing, schema inference, data validation, and background processing using BullMQ workers.
- Developed interactive ECharts visualizations and drag-and-drop dashboards with query building, filtering, chart configuration, and reusable dashboard layouts.
- Developed REST APIs with Socket.IO integration to support dataset management, real-time dashboard updates, and interactive data visualization.

Frequency-Domain Visual Prompting for Test-Time Adaptation *(Computer Vision / Prof. M Konda Reddy)*

- Developed a frequency-domain TTA method using FFT, radial frequency features, and an MLP to generate input-dependent visual prompts.
- Adapted image representations by modifying frequency amplitudes while preserving phase, with entropy minimization and L2 regularization for stable test-time updates.
- Evaluated the method on CIFAR-10-C and PACS using a frozen ResNet-18, analyzing performance under corruption and semantic domain shifts.

Autonomous Trading Agent

- Developed a Python-based trading decision system for the CLEF-2026 Financial Decision Making Task 3 using FinBERT embeddings, market features, PCA, and XGBoost for BUY/HOLD/SELL prediction.
- Evaluated the system using Combinatorial Purged Cross-Validation (CPCV), Sharpe ratio, and transaction costs, and deployed the model through a FastAPI inference API

Multispectral Satellite Image Semantic Segmentation *(Kaggle Competition)*

- Developed a Residual Attention based UNet architecture for semantic segmentation of multispectral satellite imagery as part of a Kaggle competition.
- Trained the model on 16-channel satellite data to generate pixel-wise binary segmentation masks using python.
- Evaluated and compared model performance using the Dice Coefficient, demonstrating improved performance over the standard U-Net model.

RISC-V Simulator *(Computer Architecture / Group Project / Prof. Rajesh Kedia)*

- Built an assembler and instruction-level simulator for the RISC-V ISA using C, supporting execution of core instructions.
- Integrated cache simulation with support for fully-associative, set-associative, and direct-mapped cache configurations.
- Implemented cache access tracking to generate performance metrics such as hit/miss statistics for memory operations.

TEACHING

Teaching Assistant - AI1110: Probability and Random Variables

Instructor: Dr. Anjana Mahesh

- Assisted in conducting tutorials,clarifying concepts,and supporting students with assignments.
- Helped with grading and resolving queries related to probability and random variables.

EXTRACURRICULAR

- Student Volunteer - National Service Scheme (NSS), IITH (127 hours)

- Hobbies: Chess, Volleyball, Badminton